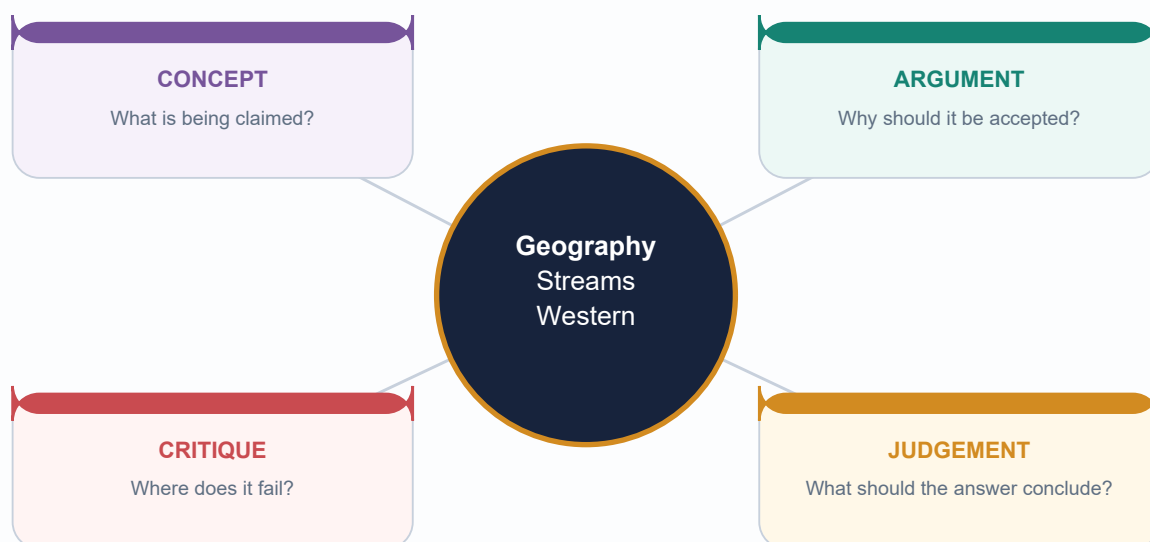


GEOGRAPHY PREMIUM SOLVED WORKBOOK

Geography 13 - Jet Streams and Western Disturbances - Solved Workbook

Official/inferred/provisional answer status preserved | Export date: 18 August 2026
36 original MCQs with strict A-B-C-D rotation | Solved 10/15/20-mark practice
Every Mains model answer ends with Why this earns marks



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

Workbook source and answer-status register

GS-I / GS-III
Geography

INTRO & ORIGIN

The workbook preserves official/inferred/provisional status and never rotates or alters actual PYQ options. Strict A -> B -> C -> D rotation applies to original MCQs only.

KEY DATA

Item	Status	Source
2020 Prelims Q99	Official paper; local official key unavailable; inferred answer labelled	CSP_2020_GS_Paper-1.pdf
2024 Prelims Q2/Q14	Official paper + official Set A key	2024-GS1-Set A.pdf; Ans-2024-GS1.pdf
2026 Prelims Q84	Official paper + provisional local key; answer labelled provisional/inferred	2026-GS1-Set A.pdf; provisional key
2021-2024 GS-I	Official question scans	Local more_previous_papers/mains folders
Original MCQs	36 questions; keys asserted A-B-C-D continuously	Generated and independently reviewed

STATIC THEORY

- PYQ wording is reproduced from local official scans in English.
- Official-key status is stated at each objective PYQ.
- Model answers use claim -> named evidence -> analysis -> qualification -> verdict and end with **Why this earns marks**.

MUST-KNOW FACTS

1. No invented PYQ status or official answer key.

UPSC TRAPS

WRONG: Original rotation is imposed on historical PYQs.

CORRECT: PYQ option order and answer are preserved; rotation applies only to original questions.

MAINS ANGLE

Use the workbook for timed attempts before reading solutions.

STUDY LINK

Local official scans and source register in the reusable Markdown.

HIGH

PYQ P1. UPSC Prelims 2020 Q99 - jet streams and cyclone eye

GS-I / GS-III
Geography

NEWS TRIGGER

Official question scan held locally. The official answer key is unavailable locally.

INTRO & ORIGIN

Consider the following statements: (1) Jet streams occur in the Northern Hemisphere only. (2) Only some cyclones develop an eye. (3) The temperature inside the eye of a cyclone is nearly 10 C lesser than that of the surroundings. Options: (A) 1 only (B) 2 and 3 only (C) 2 only (D) 3 only.

STATIC THEORY

- **INFERRED ANSWER — NOT OFFICIALLY VERIFIED: C (2 only). Confidence: High.**

- Statement 1 is false: upper-tropospheric jets occur in both hemispheres.

- Statement 2 is correct: a well-defined eye is characteristic of sufficiently organised intense tropical cyclones; not every cyclone/depression develops one.

- Statement 3 is false: the eye is a warm-core feature and is warmer aloft than the surrounding eyewall environment, not about 10 C colder.

- Elimination: rejecting 1 removes A; rejecting 3 removes B and D; C remains.

MUST-KNOW FACTS

1. PYQ status is explicit because no local official key was available.

UPSC TRAPS

WRONG: Inferred answer may be presented as official.

CORRECT: Always retain the inferred-answer label and confidence.

MAINS ANGLE

Prelims route: hemisphere -> cyclone structure -> warm-core logic.

STUDY LINK

Local scan: books\more_previous_papers\CS P_2020_GS_Paper-1.pdf, page containing Q99.

HIGH**PYQ P2. UPSC Prelims 2024 Q2 - troposphere thickness**GS-I / GS-III
Geography**NEWS TRIGGER**

Official Set A question and official Set A key are held locally.

INTRO & ORIGIN

Statement I: Thickness of the troposphere at the equator is much greater as compared to poles. Statement II: At the equator, heat is transported to great heights by strong convectional currents. Which statement combination is correct?

STATIC THEORY

- **OFFICIALLY VERIFIED ANSWER: A - both statements are correct and Statement II explains Statement I.**

- Strong tropical heating and deep convection expand the warm atmospheric column and raise the tropopause; the cold polar column is shallower.

- The question tests vertical atmosphere and causation, not a memorised altitude.

MUST-KNOW FACTS

1. Official local key: Series A, Q2 = A.

UPSC TRAPS

WRONG: Troposphere thickness is uniform globally.

CORRECT: It varies with latitude, season and temperature.

MAINS ANGLE

Prelims route: equatorial heating → deep convection → higher tropopause.

STUDY LINK

Local scans: 2024-GS1-Set A.pdf and Ans-2024-GS1.pdf.

HIGH**PYQ P3. UPSC Prelims 2024 Q14 - Coriolis force**GS-I / GS-III
Geography**NEWS TRIGGER**

Official Set A question and official Set A key are held locally.

INTRO & ORIGIN

With reference to Coriolis force: (1) It increases with increase in wind velocity. (2) It is maximum at the poles and absent at the equator. Options ask which statements are correct.

STATIC THEORY

- **OFFICIALLY VERIFIED ANSWER: C - both 1 and 2.**

- Magnitude is proportional to wind speed and the Coriolis parameter $f = 2 \Omega \sin(\text{latitude})$.

- It is zero at the equator and reaches maximum magnitude at the poles.

- Coriolis acts perpendicular to motion: right in the Northern Hemisphere, left in the Southern Hemisphere.

MUST-KNOW FACTS

1. Official local key: Series A, Q14 = C.

UPSC TRAPS

WRONG: Coriolis initiates wind from rest.

CORRECT: Pressure-gradient force initiates acceleration; Coriolis requires motion.

MAINS ANGLE

Prelims route: speed dependence + latitude dependence + direction.

STUDY LINK

Local scans: 2024-GS1-Set A.pdf and Ans-2024-GS1.pdf.

HIGH**PYQ P4. UPSC Prelims 2026 Q84 - Bharat Forecast System**GS-I / GS-III
Geography**NEWS TRIGGER**

Official question paper is held locally; only a provisional 2026 key exists locally.

INTRO & ORIGIN

Statements: (1) Bharat Forecast System aims to generate forecasts at the Panchayat-cluster level. (2) It was developed by IIT Delhi. Options: (A) 1 only (B) 2 only (C) both (D) neither.

STATIC THEORY

- **PROVISIONAL / INFERRED ANSWER — OFFICIAL UPSC KEY NOT AVAILABLE: A (1 only). Confidence: High.**
- Statement 1 is supported by official MoES/PIB descriptions of 6 km high-resolution, local/panchayat-cluster forecasting.
- Statement 2 is false: the system was developed by the Indian Institute of Tropical Meteorology (IITM), Pune, under MoES and adopted operationally by IMD.
- The local 2026 answer-key file is explicitly provisional; this package does not represent it as an official UPSC final key.

MUST-KNOW FACTS

1. BFS is a forecast model; resolution does not eliminate terrain or initial-condition uncertainty.

UPSC TRAPS

WRONG: IITM and IIT Delhi are interchangeable institutions.

CORRECT: IITM Pune developed BFS; institutional precision is examinable.

MAINS ANGLE

Prelims route: objective + developer institution.

STUDY LINK

Local 2026 paper/key plus official MoES/PIB BFS sources.

HIGH**PYQ M1. GS-I 2021 Q14 - mountain alignment and local weather**GS-I / GS-III
Geography**NEWS TRIGGER**

Official UPSC GS-I 2021 scan held locally. Directive: briefly mention and explain; 15 marks, 250 words.

INTRO & ORIGIN

Question: Briefly mention the alignment of major mountain ranges of the world and explain their impact on local weather conditions, with examples.

STATIC THEORY

- **Thesis:** Mountain alignment controls whether air is blocked, channelled or forced upward; the same range can alter precipitation, temperature, wind and storm tracks.
- **Claim - Himalaya:** East-west alignment blocks cold Central Asian surface air and forces/deflects moist flows. **Evidence:** the Himalayan-Tibetan barrier modifies the Asian jets and creates heavy orographic rain/snow on favoured slopes. **Analysis:** India has warmer winters than comparable continental latitudes and strong rain-shadow contrasts. **Qualification:** passes and synoptic systems permit episodic exchange.
- **Claim - Rockies/Andes:** Broad north-south alignment allows meridional air-mass exchange on adjacent plains while forcing Pacific westerlies upward. **Evidence:** Rocky Mountain lee cyclogenesis/Chinook and the arid Patagonian lee of the Andes. **Analysis:** windward uplift and leeward subsidence generate sharp gradients. **Qualification:** latitude and ocean current modify the result.
- **Claim - Alps:** West-east/arc alignment intercepts Atlantic-Mediterranean flow. **Evidence:** Alpine windward snowfall and Föhn warming in lee valleys. **Analysis:** relief changes phase, intensity and local temperature.
- **Verdict:** Alignment is a first-order geographical control, but local weather emerges from alignment x prevailing flow x stability x moisture x season.
- **Why this earns marks:** It answers both alignment and impact, uses named ranges/processes, compares mechanisms and ends with a qualification rather than terrain determinism.

MUST-KNOW FACTS

1. Suggested value addition: a world outline with Himalaya/Alps east-west and Rockies/Andes north-south.

UPSC TRAPS

WRONG: Only listing mountain directions answers the question.

CORRECT: The directive requires weather-process explanation with examples.

MAINS ANGLE

Claim -> range/example -> weather mechanism -> qualification -> comparative verdict.

STUDY LINK

Local official scan: QP-CSM-21-GENSTUDIE SPAPER-I-110122.pdf.

HIGH

PYQ M2. GS-I 2022 Q17 - why the troposphere determines weather

GS-I / GS-III
Geography

NEWS TRIGGER

Official UPSC GS-I 2022 scan held locally. Directive: How? 15 marks, 250 words.

INTRO & ORIGIN

Question: Troposphere is a very significant atmospheric layer that determines weather processes. How?

STATIC THEORY

- **Thesis:** The troposphere is the weather engine because it contains most atmospheric mass and water vapour, is heated largely from below and hosts the vertical/horizontal motions that convert temperature and moisture contrasts into clouds, winds and storms.

- **Claim - moisture and phase change:** Water vapour, cloud and aerosols are concentrated here. **Evidence:** condensation releases latent heat in thunderstorms, monsoon convection and tropical cyclones. **Analysis:** this couples moisture to pressure falls and circulation. **Qualification:** stratospheric variability can modulate tropospheric circulation, but weather occurs mainly below.

- **Claim - instability and convection:** Surface heating produces a normal decrease of temperature with height and buoyant overturning. **Evidence:** deep tropical convection raises the equatorial tropopause and powers monsoon clouds. **Analysis:** vertical transport creates rain, hail and lightning.

- **Claim - pressure and winds:** Horizontal temperature/pressure gradients plus Coriolis create surface winds and upper geostrophic flow. **Evidence:** subtropical jets and WDs occupy the upper troposphere; frictional convergence operates below.

- **Claim - topography/boundary layer:** Terrain, land-sea contrast and friction reshape near-surface flow. **Evidence:** Western Ghats rain shadow, Himalayan snow/rain and sea breezes.

- **Verdict:** Tropospheric depth, moisture, instability and circulation make it the immediate determinant of weather, while exchanges with the ocean, land and stratosphere condition its behaviour.

- **Why this earns marks:** It explains how rather than listing features, links microphysics to circulation and uses Indian examples plus an upper-atmosphere qualification.

MUST-KNOW FACTS

1. Diagram: surface heat/moisture → convection → upper outflow, with jet near tropopause.

UPSC TRAPS

WRONG: Weather is only a surface process.

CORRECT: Upper-tropospheric jets and troughs organise synoptic weather.

MAINS ANGLE

Structure by moisture, instability, circulation and lower boundary.

STUDY LINK

Local official scan:

QP-CSM-22-GENERAL-STUDIES-PAPER
I-190922.pdf.

HIGH**PYQ M3. GS-I 2023 Q7 - Purvaiya and Bhojpur cultural ethos**

GS-I / GS-III
Geography

NEWS TRIGGER

Official UPSC GS-I 2023 scan held locally. Directive: Why and How; 10 marks, 150 words.

INTRO & ORIGIN

Question: Why is the South-West Monsoon called Purvaiya (easterly) in Bhojpur Region? How has this directional seasonal wind system influenced the cultural ethos of the region?

STATIC THEORY

- **Thesis:** Although the all-India system is the southwest monsoon, Bhojpur commonly receives its rain-bearing branch after Bay of Bengal flow turns westward along the Ganga plain, so the experienced local wind has an easterly/southeasterly component - hence Purvaiya.

- **Claim - physical geography:** The Bay branch is deflected by the Himalayan barrier and pressure pattern along the plains. **Evidence:** east-to-west advance of rain-bearing flow across Bihar/eastern Uttar Pradesh. **Analysis:** local naming follows felt direction, not the oceanic origin label. **Qualification:** daily wind varies with monsoon trough and depressions.

- **Claim - agrarian culture:** Purvaiya marks sowing, transplanting, relief from summer heat and river/soil renewal. **Evidence:** paddy-centred farm calendars and monsoon folk traditions such as Kajri across eastern UP-Bihar. **Analysis:** wind becomes a seasonal symbol of fertility, longing, travel and reunion. **Qualification:** cultural zones overlap and modern irrigation weakens but does not erase dependence.

- **Verdict:** Purvaiya shows how a planetary monsoon is translated by regional topography into a local direction and then into language, livelihood and memory.

- **Why this earns marks:** It resolves the apparent directional paradox, uses a map-ready Bay-branch explanation and links physical geography to named cultural expression with qualification.

MUST-KNOW FACTS

1. A one-line Ganga-plain arrow from east to west is the best value addition.

UPSC TRAPS

WRONG: Southwest monsoon must blow southwest-to-northeast everywhere.

CORRECT: Branching, deflection and regional pressure fields alter local direction.

MAINS ANGLE

Physical explanation first; cultural ethos second; integrate in verdict.

STUDY LINK

Local official scan: QP-CSM-23-GENERAL-STUDIES-PAPER-I-180923.pdf.

HIGH

PYQ M4. GS-I 2024 Q6 - phenomenon of cloudbursts

GS-I / GS-III
Geography

NEWS TRIGGER

Official UPSC GS-I 2024 scan held locally. Directive: Explain; 10 marks, 150 words. This is an adjacent hazard PYQ, not a claim that every WD extreme is a cloudburst.

INTRO & ORIGIN

Question: Explain the phenomenon of cloudbursts.

STATIC THEORY

- **Thesis:** A cloudburst is exceptionally intense rain concentrated over a very small area and short duration; IMD operational teaching commonly uses about 100 mm or more in an hour over a small area.
- **Claim - formation:** Deep moist convection produces powerful updrafts and rapid condensation. **Evidence:** Himalayan relief can repeatedly lift moisture into a near-stationary cell. **Analysis:** high rain rate overwhelms infiltration and channel capacity. **Qualification:** orography favours but is not sufficient; instability and moisture convergence are required.
- **Claim - disaster conversion:** Short steep catchments concentrate runoff into flash flood/debris flow. **Evidence:** the 2013 Uttarakhand disaster involved multiple cloudburst/flash-flood/landslide processes; it should not be reduced to one label. **Analysis:** settlement on valley floors and fans multiplies loss.
- **Claim - forecast limit:** Cells are small and short-lived. **Evidence:** radar/AWS/nowcasting improve detection but mountain beam blockage and sparse stations persist. **Analysis:** land-use control and local warning are as important as model improvement.
- **Verdict:** Cloudburst risk is the interaction of extreme rain rate with Himalayan relief and exposure, not simply 'heavy monsoon rain'.
- **Why this earns marks:** It defines the threshold logic, explains microphysics and terrain, names a qualified Indian example and includes forecast/land-use limits.

MUST-KNOW FACTS

1. Terminology caution is central to this package.

UPSC TRAPS

- WRONG:** Every heavy WD rain spell is a cloudburst.
- CORRECT:** Use the label only with suitable high-resolution rainfall evidence.

MAINS ANGLE

Definition -> mechanism -> terrain/runoff -> forecasting/mitigation.

STUDY LINK

Local official scan: UPSC Mains 2024 GS Paper I.pdf.

HIGH

Hard MCQ set 1-2

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY**- Q1. Which description best defines a jet stream?**

- A. A narrow elongated wind-speed maximum near the upper troposphere/tropopause
- B. Any wind faster than local surface winds
- C. A permanent current at exactly 12 km
- D. A low-level moisture plume only

- **Key: A** - A is correct. Jets are relative elongated speed maxima; level and speed vary.

- **Trap checked:** Fixed altitude/speed definition.

- Q2. The thermal wind is best understood as:

- A. A hot downslope wind
- B. Vertical shear of geostrophic wind linked to a horizontal temperature gradient
- C. Surface wind across isotherms
- D. Frictional inflow toward a low

- **Key: B** - B is correct. Thermal wind is a vector shear relation, not a separately observed local wind.

- **Trap checked:** Name-based confusion.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 3-4**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q3. On a constant-pressure chart, tightly packed geopotential-height contours most directly indicate:**

- A. High humidity
- B. Strong vertical motion
- C. Strong horizontal height gradient and potentially strong geostrophic wind
- D. Low surface elevation

- **Key: C** - C is correct. Wind strength follows the height gradient; ascent needs additional diagnosis.

- **Trap checked:** Equating contour spacing with rainfall.

- **Q4. Why can gradient wind differ from geostrophic wind?**

- A. Humidity changes Coriolis sign
- B. Pressure gradient disappears
- C. Friction is always dominant aloft
- D. Curvature adds a centrifugal/curvature term

- **Key: D** - D is correct. Curved flow requires a three-term balance.

- **Trap checked:** Applying straight-flow balance to curved jets.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 5-6**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q5. Which statement correctly contrasts the polar-front and subtropical jets?**

- A. The former is strongly eddy/baroclinic; the latter also reflects upper Hadley-cell angular momentum
- B. Both are surface winds
- C. Only the subtropical jet occurs in winter
- D. The polar-front jet is an easterly

- **Key: A** - A is correct; both are upper westerly currents with different dominant dynamical settings.

- **Trap checked:** Treating them as identical lines.

- **Q6. Rossby-wave meanders arise fundamentally from:**

- A. Ocean tides
- B. Latitudinal variation of planetary vorticity and potential-vorticity dynamics
- C. Local mountain breezes
- D. Surface friction alone

- **Key: B** - B is correct. Relief and heating can anchor waves, but beta/PV dynamics are fundamental.

- **Trap checked:** Calling them ocean waves.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 7-8**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q7. For a straight west-to-east Northern Hemisphere jet streak, the first-order divergence sectors are:**

- A. Left entrance and right exit
- B. Both entrance sectors
- C. Right entrance and left exit
- D. Both exit sectors

- **Key: C** - C is correct, with the major caveat that curvature and deformation modify the rule.

- **Trap checked:** Mechanical use in curved flow.

- **Q8. Which current is correctly identified as a lower-tropospheric moisture-bearing jet?**

- A. STWJ
- B. TEJ
- C. Polar-front jet
- D. Somali/Findlater jet

- **Key: D** - D is correct. It is the cross-equatorial monsoon low-level jet.

- **Trap checked:** Assuming every named jet is upper-level.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 9-10**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q9. The winter STWJ is most directly relevant to India because it:**

- A. Provides an upper westerly steering environment for WDs
- B. Carries Arabian Sea moisture at low level
- C. Is the same as TEJ
- D. Creates every cold wave

- **Key: A** - A is correct. Steering does not mean it alone determines rainfall or cold waves.

- **Trap checked:** Conveyor-belt determinism.

- **Q10. The TEJ over South Asia is best treated as:**

- A. A winter westerly
- B. A summer upper easterly associated with monsoon heating and the Tibetan anticyclone
- C. A surface sea breeze
- D. The only cause of onset

- **Key: B** - B is correct. It is both part of and responsive to established monsoon heating.

- **Trap checked:** Single-cause onset story.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 11-12**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q11. Which wording is most scientifically defensible?**

- A. STWJ withdrawal causes onset by itself
- B. TEJ always appears before convection
- C. Northward STWJ reorganisation is associated with onset but acts with low-level flow, heating and convection
- D. ENSO decides onset date exactly

- **Key: C** - C is correct because it preserves the multicausal, coupled nature of onset.

- **Trap checked:** Deterministic verbs.

- **Q12. The Tibetan anticyclone is primarily:**

- A. A surface winter high over Siberia
- B. A Western Disturbance
- C. The Somali jet
- D. An upper-level summer anticyclonic circulation linked to deep monsoon heating

- **Key: D** - D is correct; TEJ lies broadly on its southern flank.

- **Trap checked:** Confusing upper high with surface thermal low.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 13-14**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q13. Which is the most precise definition of a Western Disturbance?**

- A. An eastward-moving synoptic extratropical disturbance in subtropical westerlies, often an upper trough/circulation
- B. Every winter surface low over Delhi
- C. A Bay of Bengal monsoon depression
- D. A local hailstorm

- **Key: A** - A is correct and leaves room for an induced lower circulation without requiring a strong surface cyclone.

- **Trap checked:** Surface-low-only definition.

- **Q14. Which statement about WD moisture is correct?**

- A. It always comes only from the Mediterranean
- B. Sources are event-specific and may include Mediterranean/West Asian seas, Arabian Sea or regional pathways
- C. Moisture source and steering current are identical
- D. WDs require Bay of Bengal moisture

- **Key: B** - B is correct. Track shorthand must not become parcel-source certainty.

- **Trap checked:** Single-source universalisation.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 15-16**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q15. An induced cyclonic circulation associated with a WD most commonly refers to:**

- A. The TEJ core
- B. A tropical cyclone eye
- C. A lower-tropospheric cyclonic response that can organise convergence over Pakistan/NW India
- D. A polar high

- **Key: C** - C is correct. It is related to but distinct from the upper WD trough.

- **Trap checked:** Calling upper and lower centres identical.

- **Q16. Which combination best explains heavy WD precipitation over the western Himalaya?**

- A. Jet speed only
- B. Surface temperature only
- C. Mediterranean origin only
- D. Dynamic ascent + moisture + lower convergence + orographic lift

- **Key: D** - D is correct; precipitation efficiency is multicomponent.

- **Trap checked:** One-factor explanation.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 17-18**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q17. Which agriculture statement is correct?**

- A. WD rain can help Rabi crops when moderate and timely but can damage through hail, lodging or harvest rain
- B. More WD rain is always beneficial
- C. Wheat is a Kharif crop
- D. Hail affects orchards only

- **Key: A** - A is correct. Crop stage and drainage determine the sign of impact.

- **Trap checked:** Benefit-only narrative.

- **Q18. Which hazard warning is institutionally matched correctly?**

- A. DGRE - tropical cyclone naming
- B. DGRE - avalanche danger bulletins
- C. CWC - upper-air jet analysis
- D. IMD - glacier inventory

- **Key: B** - B is correct. IMD weather and CWC hydrology have different roles.

- **Trap checked:** One-agency Himalayan hazard assumption.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 19-20**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q19. Which feature most clearly distinguishes a WD from a monsoon depression?**

- A. Both can produce rain
- B. Both may show cyclonic circulation
- C. WD is an extratropical upper-westerly disturbance; monsoon depression is embedded in tropical monsoon flow
- D. Both always move east

- **Key: C** - C is correct. Level, energy source, season and track complete the distinction.

- **Trap checked:** Low-pressure synonym trap.

- **Q20. A robust WD climatology should avoid:**

- A. Separating track from intensity
- B. Naming the tracking method
- C. Stating the season
- D. Treating frequency, intensity and precipitation as one interchangeable metric

- **Key: D** - D is correct. Different metrics can trend differently.

- **Trap checked:** Single-number climatology.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 21-22**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY**- Q21. Which climate-change statement is most defensible?**

- A. Moisture loading can intensify precipitation even if WD circulation/frequency does not increase
- B. Every WD is now stronger
- C. One wet winter proves a trend
- D. Snowfall and precipitation are identical

- **Key: A** - A is correct; thermodynamic and dynamic changes must be separated.

- **Trap checked:** Universal metric claim.

- Q22. Which sequence correctly separates operational products?

- A. Warning -> diagnosis -> observation
- B. Diagnosis/status -> forecast -> warning/action -> verification
- C. Reanalysis -> outcome -> forecast
- D. Impact -> model initialisation -> satellite

- **Key: B** - B is correct. Products differ by time orientation and purpose.

- **Trap checked:** Treating forecast as outcome.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 23-24**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q23. ERA5 is best described as:**

- A. A rain gauge network
- B. An IMD warning
- C. A retrospective model-data reanalysis used for diagnosis
- D. A deterministic future forecast only

- **Key: C** - C is correct. It assimilates observations but is not direct observation everywhere.

- **Trap checked:** Reanalysis-as-observation.

- **Q24. A DGRE avalanche danger level should be interpreted as:**

- A. A guarantee for every slope
- B. A snowfall forecast only
- C. A district crop advisory
- D. A time-bounded hazard assessment based on snow/met data and projected weather, with residual uncertainty

- **Key: D** - D is correct and reflects DGRE's own disclaimer.

- **Trap checked:** Guarantee interpretation.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 25-26**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q25. In the IMD 19 March 2026 release, the valid 19-20 March rain/snow statement is:**

- A. A forecast, not proof of later outcome
- B. A climate trend
- C. A seasonal verification
- D. A glacier assessment

- **Key: A** - A is correct. Agency/date/validity must accompany the claim.

- **Trap checked:** Forecast-outcome conflation.

- **Q26. The WMO July 2026 HiM-DATA item is most safely used as:**

- A. Proof India has complete high-altitude coverage
- B. Regional evidence that high-Himalayan observation gaps remain and community AWS can help
- C. A WD forecast for India
- D. An avalanche warning

- **Key: B** - B is correct; the initiative discussed Nepal/Bhutan and is not an Indian operational-status claim.

- **Trap checked:** Geographic overextension.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Hard MCQ set 27-28**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q27. Bharat Forecast System was developed by:**

- A. IIT Delhi
- B. DGRE
- C. IITM Pune under MoES
- D. CWC

- **Key: C** - C is correct. IMD adopted it operationally.

- **Trap checked:** Institutional-name confusion.

- **Q28. When is 'cloudburst' terminology justified?**

- A. Any winter rain
- B. Any forecast of heavy rain
- C. Any WD over mountains
- D. When very high hourly rain is observed over a small area consistent with the operational criterion

- **Key: D** - D is correct. Use 'intense/heavy rain' when high-resolution evidence is absent.

- **Trap checked:** Decorative hazard language.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Remedial MCQ set 1-2**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q1. A student writes: 'Thermal wind blows from warm to cold air.' The best correction is:**

- A. Thermal wind is the vertical geostrophic-wind shear implied by the horizontal temperature gradient
- B. It is an anabatic wind
- C. It is the TEJ
- D. The sentence is fully correct

- **Key: A** - A corrects the conceptual category error.

- **Trap checked:** Literal reading of 'wind'.

- **Q2. A strong WD produces little precipitation over one basin. The best explanation is:**

- A. Impossible by definition
- B. Track/moisture/stability or terrain may make the system precipitation-inefficient there
- C. All WDs carry equal moisture
- D. Jet speed determines rain exactly

- **Key: B** - B separates dynamical strength from precipitation efficiency.

- **Trap checked:** Strength-rain equivalence.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Remedial MCQ set 3-4**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q3. A forecast map shows a 60% chance of heavy snow. Which statement is correct?**

- A. Heavy snow definitely occurred
- B. No warning is possible
- C. It expresses conditional probability for the stated place/time and needs later verification
- D. It is a climate projection

- **Key: C** - C demonstrates forecast literacy.

- **Trap checked:** Probability-as-certainty.

- **Q4. After a WD passes, clear skies and light winds can sometimes favour:**

- A. TEJ formation
- B. Tropical cyclone genesis
- C. Permanent warming
- D. Radiational cooling, frost/fog or cold-wave intensification depending on air mass

- **Key: D** - D is correct; temperature response changes across event phases.

- **Trap checked:** One-temperature-response assumption.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Remedial MCQ set 5-6**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q5. Which answer sentence is best?**

- A. The STWJ shift is an onset association within a coupled monsoon reorganisation
- B. The STWJ alone causes onset
- C. TEJ is a winter jet
- D. Somali jet is upper easterly

- **Key: A** - A uses bounded causal language.

- **Trap checked:** Textbook determinism.

- **Q6. A district has more rain but fewer counted WDs in a decade. This is:**

- A. Logically impossible
- B. Possible if precipitation efficiency, track, duration or moisture increased
- C. Proof the data are false
- D. Proof of more surface cyclones

- **Key: B** - B is correct because metrics can diverge.

- **Trap checked:** Metric conflation.

MUST-KNOW FACTS

1. Rotation keys in this card: A, B.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Remedial MCQ set 7-8**

GS-I / GS-III
Geography

INTRO & ORIGIN

Original question set. Correct options continue the strict A -> B -> C -> D rotation.

STATIC THEORY

- **Q7. Which qualification belongs after citing a jet-streak quadrant?**

- A. No caveat is needed
- B. It applies only in Southern Hemisphere
- C. Curvature, shear and ageostrophic structure can modify the straight-jet rule
- D. It predicts exact rainfall

- **Key: C** - C is the required scientific caveat.

- **Trap checked:** Mechanical quadrant use.

- **Q8. DGRE closes regular avalanche bulletins for summer. The best inference is:**

- A. No snow remains anywhere
- B. Climate change ended avalanches
- C. IMD stops weather forecasts
- D. Seasonal danger reduced enough to close routine service, while residual loaded slopes still require caution

- **Key: D** - D matches the 1 June 2026 official notice.

- **Trap checked:** Service closure-as-zero-risk.

MUST-KNOW FACTS

1. Rotation keys in this card: C, D.

UPSC TRAPS

WRONG: An option is made obviously longer or more qualified than others.

CORRECT: Options were independently checked for plausibility and statement accuracy.

MAINS ANGLE

After answering, explain why each rejected option fails; do not rely on the key letter.

STUDY LINK

Coverage follows the complete teaching sequence.

HIGH**Original Mains 10 marks - upper jets versus Somali low-level jet**GS-I / GS-III
Geography**INTRO & ORIGIN**

Question: Distinguish the Subtropical Westerly Jet, Tropical Easterly Jet and Somali/Findlater low-level jet, and explain their seasonal relevance to India. Answer in 150 words.

STATIC THEORY

- **Thesis:** The three are not variants of one wind: they differ in level, direction, dynamical setting and seasonal function.
- **Claim - STWJ:** It is an upper westerly associated with subtropical temperature/momentum gradients. **Evidence:** in winter its Asian branch commonly lies south of the Himalaya and steers WDs. **Analysis:** it links mid-latitude waves to north-Indian winter weather. **Qualification:** position varies daily and it does not alone determine precipitation.
- **Claim - TEJ:** It is an upper easterly associated with the summer Tibetan anticyclone and monsoon heating. **Evidence:** it becomes prominent during established monsoon circulation. **Analysis:** it forms part of upper outflow/shear. **Qualification:** it is not the sole onset switch.
- **Claim - Somali LLJ:** It is lower-tropospheric cross-equatorial southwesterly flow. **Evidence:** it transports Arabian Sea moisture toward India. **Analysis:** it supplies low-level moisture and momentum. **Qualification:** rainfall still needs monsoon trough, convection and relief.
- **Verdict:** Level-direction-season labelling prevents the common error of equating every 'jet' with an upper westerly.
- **Why this earns marks:** It answers the directive comparatively, uses India-specific evidence, explains function and qualifies causation within the word-limit logic.

MUST-KNOW FACTS

1. Add a three-arrow vertical sketch for value.

UPSC TRAPS

WRONG: Narrating the three jets without comparison.

CORRECT: Use level, direction, season and role as parallel headings.

MAINS ANGLE

Claim -> named jet/evidence -> analysis -> qualification -> verdict.

STUDY LINK

Visual 05 and distinction matrix.

HIGH

Original Mains 15 marks - pathway and impacts of Western Disturbances

GS-I / GS-III
Geography

INTRO & ORIGIN

Question: Explain the pathway and precipitation mechanism of Western Disturbances and assess why the statement 'they originate over the Mediterranean and benefit Indian agriculture' is incomplete. Answer in 250 words.

STATIC THEORY

- **Thesis:** WDs are eastward-moving extratropical disturbances in subtropical westerlies; 'Mediterranean origin' and 'agricultural benefit' are useful entry points but incomplete generalisations.

- **Claim - pathway:** Baroclinic troughs/circulations develop or amplify across the Mediterranean-West Asian-Central Asian corridor. **Evidence:** IMD operationally analyses them as upper troughs/cyclonic circulations, sometimes with induced lower circulations over Pakistan-NW India. **Analysis:** the STWJ/Rossby pattern steers rather than supplies all moisture.

Qualification: individual genesis and tracks differ.

- **Claim - moisture/ascent:** Precipitation needs upper forcing, lower convergence, moisture and Himalayan lift. **Evidence:** MAUSAM's 26-29 December 2024 case found active subtropical-jet divergence, WD-easterly coupling and Arabian Sea plus Bay moisture. **Analysis:** nearby moisture can dominate event efficiency. **Qualification:** this rare coupled case is not universal.

- **Claim - benefits:** Moderate winter rain can support Rabi soil moisture and snowfall stores water. **Evidence:** Punjab-Haryana-western UP wheat belt and western-Himalayan snow dependence. **Analysis:** timing complements irrigation/residual moisture. **Qualification:** hail, lodging, disease, waterlogging, avalanche and road disruption can reverse the sign.

- **Verdict:** WDs should be analysed as variable upper-air systems whose outcome is controlled by track x moisture x terrain x exposure, not by origin or benefit slogans.

- **Why this earns marks:** It corrects both halves of the statement, names official/scientific evidence, integrates physical and human geography and gives a conditional verdict.

MUST-KNOW FACTS

1. Map: corridor + separate moisture arrows + Himalaya.

UPSC TRAPS

WRONG: Using the December 2024 coupled case as universal climatology.

CORRECT: Retain the event-specific qualification.

MAINS ANGLE

Definition -> track -> ascent/moisture -> impacts -> qualification -> verdict.

STUDY LINK

Visuals 07-10; IMD/MAUSAM current evidence.

HIGH

Original Mains 20 marks - climate change and forecast governance

GS-I / GS-III
Geography

INTRO & ORIGIN

Question: Critically assess how climate change may alter Western-Disturbance impacts over India, and evaluate whether improved forecasting alone can manage the resulting risk. Answer in about 300 words.

STATIC THEORY

- **Thesis:** Climate change is more likely to alter moisture loading, rain-snow partition and extremes than to produce one uniform change in WD frequency; forecast improvement is necessary but insufficient without terrain observations, impact translation and exposure reduction.
- **Claim - thermodynamics:** Warmer air supports more water vapour and a higher freezing level. **Evidence:** the WCD review synthesises experiments where increased humidity raised WD precipitation despite weakened circulation. **Analysis:** lower elevations may receive more rain while extreme precipitation potential rises. **Qualification:** elevation, season and model produce heterogeneous outcomes.
- **Claim - dynamics:** Jet position, baroclinicity, blocking and WD track may change. **Evidence:** literature reviewed in WCD reports divergent frequency/track results and possible changing seasonality. **Analysis:** dynamic uncertainty can offset or reinforce thermodynamic change. **Qualification:** one winter or bulletin cannot establish trend/attribution.
- **Claim - risk:** More intense rain, wet snow or rapid phase shifts can cascade into avalanche, landslide, flood, crop loss and transport disruption. **Evidence:** DGRE's snowpack-based warning and IMD's 19 March 2026 hail/snow impact advice show hazard-specific pathways. **Analysis:** exposure converts weather into disaster.
- **Claim - forecast governance:** Mission Mausam/BFS, radiosondes, radar, satellites and ensembles improve diagnosis/lead time. **Evidence:** BFS is IITM's 6 km operational system; WMO July 2026 still notes high-altitude observational gaps regionally. **Analysis:** small track errors remain costly in steep terrain. **Qualification:** model resolution is not exact local truth.
- **Way forward:** denser mountain AWS/snow gauges, ensemble impact thresholds, IMD-DGRE-CWC-State interoperability, protected avalanche/flood corridors, crop advisories, drainage and verified warning performance.
- **Verdict:** Resilience requires forecast skill x institutional coordination x land-use/crop/route action; none can substitute for the others.
- **Why this earns marks:** It separates thermodynamic/dynamic evidence, uses named current institutions, evaluates the 'forecast alone' proposition and ends with a graded governance verdict.

MUST-KNOW FACTS

1. Use visual 12 for climate logic and visual 11 for forecast governance.

UPSC TRAPS

WRONG: Claiming a settled increase in every WD metric.

CORRECT: State where evidence is robust and where it remains uncertain.

MAINS ANGLE

Critical structure: evidence for -> uncertainty/counterpoint -> governance evaluation -> integrated verdict.

STUDY LINK

WCD 2025, IMD/DGRE 2026, Mission Mausam/BFS and WMO observation context.

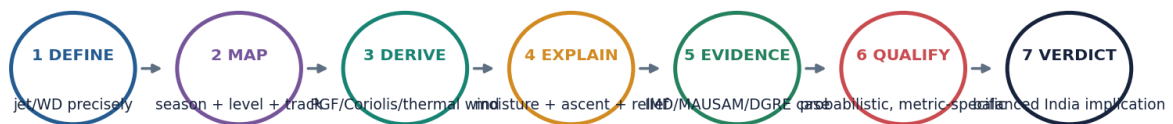
HIGH

Examiner-grade answer checklist and self-review rubric

GS-I / GS-III
Geography

EXAMINER-GRADE ANSWER CHECKLIST AND SELF-REVIEW RUBRIC - VISUAL

UPSC answer spine for jet streams and Western Disturbances



Every model answer: claim -> named evidence/example -> analysis -> qualification -> verdict

Apply this sequence before comparing with the model solution.

AI-generated illustration - conceptual visual, not factual evidence.

KEY DATA

Criterion	Self-check
Demand fidelity	Every directive, scope, mark and word limit answered
Evidence	2-3 named examples (10m), 4-6 (15m), 5-8 (20m) where appropriate
Analysis	Each example states what it proves
Qualification	Event/trend/forecast uncertainty bounded
Structure	Map/diagram/table used where it improves clarity
Verdict	Direct, graded and derived from body
10-mark timing	About 7 minutes: 1-line frame, 3 analytical blocks, qualified verdict
15-mark timing	About 11 minutes: compact diagram plus 4-5 linked dimensions
20-mark timing	About 15 minutes: define, explain mechanism, assess impacts, qualify and conclude
Revision log	Record the missing mechanism, wrong level/season, weak evidence and rewritten verdict

STATIC THEORY

- Do not award yourself full credit for correct facts arranged as a list.
- Underline the metric and validity window in every current-affairs sentence.
- For diagrams, check arrow direction, atmospheric level, season and caveat.
- For climate questions, separate thermodynamic and dynamic change.
- For agriculture/disaster, write conditional impact rather than one-direction benefit/loss.
- A jet/WD answer should normally distinguish diagnosis, forecast, warning and verified outcome.
- A map should separate the steering corridor from event-specific moisture-source arrows.
- A mechanism paragraph should link forcing -> ascent/convergence -> cloud/precipitation -> conditional impact.
- A comparison answer should explicitly distinguish STWJ, TEJ, Somali LLJ, WD, monsoon depression and tropical cyclone where relevant.
- After review, rewrite only the weakest paragraph and the verdict; do not merely reread the model answer.

MUST-KNOW FACTS

1. Internal review weighting: demand 20; accuracy/evidence 30; analysis 20; structure 15; balance/conclusion 10; presentation 5.
2. Minimum evidence audit: one named official/scientific source, one India-specific spatial example and one explicit qualification where the question permits.
3. Diagram audit: title, orientation/level, labelled arrows, process link and one caveat.

UPSC TRAPS

WRONG: Named evidence is decorative.

CORRECT: Every example must advance or qualify an argument.

WRONG: A forecast is cited as an observed disaster.

CORRECT: Retain issue date, validity window and FORECAST label; verify outcomes separately.

WRONG: One winter proves a long-term WD trend.

CORRECT: Use metric-specific multi-year evidence and state uncertainty.

MAINS ANGLE

Rewrite any answer lacking a named source/example or a LIMIT sentence.

STUDY LINK

Repository Mains model-answer standard in AGENT_MEMORY.md.

End of Register Notes - UPSC Agent / Copilot CLI